

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Image Upload & Classification	Upload butterfly image - Preprocess image (resize, normalize)
FR-4	Result Visualization	Display predicted species and confidence score - Show educational info about species
FR-5	Dashboard	streamlit-based interface for image upload and prediction - Reset / Clear input
FR-6	Admin Functions - Add new data for training - Trigger model retraining	Admin Functions - Add new data for training - Trigger model retraining

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide a clean, intuitive interface accessible to all users.
NFR-2	Security	Data transmission and image uploads must be encrypted using HTTPS protocols.
NFR-3	Reliability	The prediction model should return consistent results
NFR-4	Performance	

		Image classification should be completed in less than 3 seconds or
NFR-5	Availability	The app should be available 24/7 with uptime of 99% during demo week.
NFR-6	Scalability	The architecture should support adding more species